

**Kyle Gontjes, PhD, MPH**  
**Research Investigator**  
**Department of Microbiology & Immunology**  
**University of Michigan Medical School**  
**1150 W. Medical Center Dr.**  
**1510A Medical Science Research Building I**  
**(616) 617-7843 - kgontjes@umich.edu**

## **Education and Training**

### **Education**

|                 |                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------|
| 09/2014-12/2017 | BSc, Biology, Calvin University, Grand Rapids, United States                                                     |
| 09/2018-05/2020 | MPH, Hospital & Molecular Epidemiology, University of Michigan–Ann Arbor, Ann Arbor, United States               |
| 09/2020-05/2026 | PhD, Microbiology & Immunology, Microbiology & Immunology, University of Michigan–Ann Arbor, Ann Arbor, Michigan |

### **Postdoctoral Training**

|                 |                                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 05/2026-08/2026 | Postdoctoral Fellow, Genomic epidemiology, University of Michigan Medical School, Microbiology & Immunology, Ann Arbor, MI, Evan Snitkin |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|

## **Work Experience**

### **Research Position**

|                 |                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| 05/2018-09/2020 | Research Assistant, Laboratory of Dr. Lona Mody, Internal Medicine, University of Michigan Medical School, Ann Arbor, MI |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|

## **Research Interests**

- Genomic epidemiology of microbial pathogens
- Microbial evolution
- Antibiotic resistance
- Healthcare epidemiology and infection prevention
- Method development for genomic epidemiology

## **Grants**

### **Submitted - Open**

R01:*DEterminants of ESCRE Colonization and Transmission in the Community (DETECT):*  
Co-I (Principal Investigator:Evan Snitkin)  
NIH-DHHS-US-SubK sourced funding through The University of Pennsylvania  
04/2027 - 03/2032  
\$954,826

*Early-Life Resistance Networks: Amplification and Transmission of ESBL Plasmids in Children and Households:*

Co-I (Principal Investigator:Evan Snitkin)  
Wellcome Leap Inc sourced funding through Ann and Robert H. Lurie Childr

11/2026 - 10/2029  
\$188,964

### **Past Grants**

F31AI186288: *Genome-driven identification of patient and bacterial drivers of resistance to novel beta-lactam/beta-lactamase inhibitor combination therapies:*

PI

NIH-DHHS-US

08/2024 - 07/2026

\$84,318

## **Honors and Awards**

### **National**

2020 - Present      SHEA Yoshikawa-High Award, Society of Healthcare Epidemiology of America, United States

### **Regional**

2025 - Present      NIH Trainee Award, Midwest Microbial Pathogenesis Conference, United States

### **Institutional**

2025 - Present      University of Michigan Rackham Travel Award, University of Michigan, United States

### **Departmental**

2025 - Present      Frederick Novy Award, University of Michigan, Microbiology & Immunology, United States

## **Study Sections, Editorial Boards, Journal & Abstract Review**

### **Editorial Boards / Journal & Abstract Reviews**

#### **Journal Review**

2024 - Present      Antimicrobial Resistance and Infection Control (Ad Hoc)  
2025 - Present      Scientific Reports (Ad Hoc)  
2025 - Present      BMC Infectious Diseases (Ad Hoc)  
2025 - Present      Journal of Infectious Diseases (Ad Hoc)  
2026 - Present      Genome Medicine (Ad Hoc)

## **Teaching**

### **Mentorship**

#### **Undergraduate Student**

10/2023 - Present      Aryan Singh, University of Michigan, Working in lab

### **Teaching Activity**

#### **Institutional**

12/2023 - 04/2025      Assisted in class discussions, laboratory assistance, and grading for EPID 582 Molecular Epidemiology & Pathogen Genomics, University of Michigan School of Public Health, Epidemiology, Course Assistant

## Memberships in Professional Societies

2017 - Present                      Member, ASM

## Committee/Service

### Departmental

2020 - 2021                      Organization of Microbiology & Immunology Students, University of Michigan Medical School, Other, Co-Social Chair

## Scholarly Activities

### Presentations

#### Extramural Invited Presentation

##### Seminar

1. The Making of A Disease Detective: A Journey in Healthcare & Molecular Epidemiology, **Gontjes K**, Calvin University, 02/2023, Grand Rapids, Michigan
2. Application of Whole-Genome Sequencing to the Study of Antibiotic Resistance, Gontjes K, Calvin University, 09/2024, Grand Rapids, Michigan

#### Intramural Invited Presentation

##### Seminar

1. Contamination of Common Area and Rehabilitation Gym Environment with Multidrug-Resistant Organisms, **Gontjes K**, University of Michigan Division of Geriatric & Palliative Medicines Clinical Epidemiology Conference, 01/2019, Ann Arbor, MI
2. Can Environmental Surveillance Replace Perianal Screening?, **Gontjes K**, University of Michigan Division of Geriatric & Palliative Medicine's Clinical Epidemiology Conference, 01/2020, Ann Arbor, MI

## Publications/Scholarship

(Co-First Author \*; Corresponding author \*\*; Co-Last author \*\*\*)

### Peer Reviewed

#### Journal Article

1. **Gontjes KJ**, Gibson KE, Lansing B, Cassone M, Mody L: Contamination of Common Area and Rehabilitation Gym Environment with Multidrug-Resistant Organisms. *J Am Geriatr Soc*.68(3): 478-485, 03/2020. PM31851386
2. **Gontjes KJ**, Gibson KE, Lansing BJ, Mody L, Cassone M: Assessment of race and sex as risk factors for colonization with multidrug-resistant organisms in six nursing homes. *Infect Control Hosp Epidemiol*.41(10): 1222-1224, 10/2020. PM32493534
3. Cassone M, Mantey J, **Gontjes KJ**, Lansing BJ, Gibson KE, Wang J, Mody L: Seasonal Patterns in Incidence and Antimicrobial Resistance of Common Bacterial Pathogens in Nursing Home Patients and Their Rooms. *Front Public Health*.9: 671428, 01/2021. PM34322470
4. Mody L, **Gontjes KJ**, Cassone M, Gibson KE, Lansing BJ, Mantey J, Kabeto M, Galecki A, Min L: Effectiveness of a Multicomponent Intervention to Reduce Multidrug-Resistant Organisms in Nursing Homes: A Cluster Randomized Clinical Trial. *JAMA Netw Open*.4(7): e2116555, 07/2021. PM34269807
5. **Gontjes KJ**, Gibson KE, Lansing BJ, Mantey J, Jones KM, Cassone M, Wang J, Mills JP, Mody L, Patel PK: Association of Exposure to High-risk Antibiotics in Acute Care Hospitals With Multidrug-Resistant Organism Burden in Nursing Homes. *JAMA network open*.5(2)01/2022. PM35103795
6. **Gontjes KJ**, Gibson KE, Lansing B, Cassone M, Mody L: Can alternative anatomical sites and environmental surveillance replace perianal screening for multidrug-resistant organisms in nursing homes?.

*Infect Control Hosp Epidemiol.*43(8): 1063-1066, 08/2022. PM34016196

7. Wolfensberger A, Mang N, Gibson KE, **Gontjes K**, Cassone M, Brugger SD, Mody L, Sax H: Understanding short-term transmission dynamics of methicillin-resistant *Staphylococcus aureus* in the patient room. *Infect Control Hosp Epidemiol.*43(9): 1147-1154, 09/2022. PM34448445
8. Mody M, **Gontjes KJ**: Health education after COVID-19: A time to revisit, revamp, and revitalize?. *Am J Infect Control.*51(4): 466-468, 04/2023. PM36442688
9. Zhang HL, **Gontjes KJ**, Han JH, Alby K, Lapp Z, Snitkin E, Goldstein EJ C, Muldoon S, Tolomeo P, Lautenbach E: Characterization of resistance to newer antimicrobials among carbapenem-resistant *Klebsiella pneumoniae* in the post-acute-care setting. *Infect Control Hosp Epidemiol.*44(7): 1159-1162, 07/2023. PM35896514
10. Cassone M, Wang J, Lansing BJ, Mantey J, Gibson KE, **Gontjes KJ**, Mody L: Diversity and persistence of MRSA and VRE in skilled nursing facilities: environmental screening, whole-genome sequencing and development of a dispersion index. *J Hosp Infect.*138: 8-18, 08/2023. PM37160232
11. Lapp Z, Octaria R, O'Malley SM, Nguyen TN, Wolford H, Crawford R, Moore C, Snippes Vagnone P, Noel D, Duffy N, Pirani A, Thomas LS, Pattee B, Pearson C, Bulens SN, Hoffman S, Kainer M, Anacker M, Meek J, See I, **Gontjes KJ**, Chan A, Lynfield R, Maloney M, Hayden MK, Snitkin E\*\*, Slayton RB\*\*: Distinct Origins and Transmission Pathways of blaKPCEnterobacterales across Three U.S. States. *Journal of clinical microbiology.*61(8): e0025923-e0025923, 08/2023. PM37439675
12. Mody M, Hodges AN, Gibson KE, Mantey J, **Gontjes KJ**: Does high school health education class satisfaction influence student perceptions of public health and pandemic response?. *Am J Infect Control.*52(6): 719-725, 06/2024. PM38253288
13. Unverdorben LV, Pirani A, **Gontjes K**, Moricz B, Holmes CL, Snitkin ES, Bachman MA: *Klebsiella pneumoniae* evolution in the gut leads to spontaneous capsule loss and decreased virulence potential. *mBio.*16(5): e0236224, 05/2025. PM40162782
14. Thiede SN, Steinberg H, Benedict E, Sansom S, Aroutcheva A, **Gontjes KJ**, Winner K, Royzenblat S, Pirani A, Weinstein RA, Zawitz C, Moore NM, Popovich KJ, Snitkin ES: Antibiotic-resistance plasmid amplified among MRSA cases in an urban jail and its connected communities. *Nat Commun.*17(1): 420, 12/2025. PM41353225
15. **Gontjes KJ**, Singh A, Sansom SE, Boyko JD, Smith SA, Lautenbach E, Snitkin E: Phylogenetic Context of Antibiotic Resistance Provides Insights into the Dynamics of Resistance Emergence and Spread. *J Infect Dis.*232(6): e992-e1002, 12/2025. PM40971902
16. Boyko JD, **Gontjes KJ**, Snitkin ES, Smith SA: Resolving competing evolutionary histories in joint ancestral state reconstruction. *Proc Natl Acad Sci U S A.*123(15): e2528064123, 04/2026. PM41955119
17. **Gontjes KJ**, Kelly BJ, Cressman L, Jaskowiak-Barr A, Glaser L, Cowden LL, Patel A, Han J, Goldstein EJ C, Bilker WB, Snitkin ES, Lautenbach E: Changing molecular epidemiology and resistance patterns of carbapenem-resistant *Klebsiella pneumoniae* in California long-term acute care hospitals. *J Infect Dis.*08/2026. PM42566283

## Non-Peer Reviewed

### Interview

1. Gontjes K: Hospital Exposure to Antibiotics Associated with C diff Pose Higher Risk of MDRO, 10/2020. <https://www.contagionlive.com/view/hospital-exposure-to-antibiotics-associated-with-c-diff-pose-higher-risk-of-mdro>

### Letter

1. **Gontjes K**, Collier K: From a Culture of Blame to a Culture of Grace: A Letter in Reply to Papadakis. *Infection Control and Hospital Epidemiology.*: 1-2, 08/2021. PM34446119

### Meeting Abstract

1. **Gontjes K**, Gibson K, Lansing B, Cassone M, Mody L: Simplifying Surveillance Sampling: Can Environmental Surveillance Replace Perianal Screening?. *Infection Control and Hospital*

*Epidemiology*.41(S1): s15-s16, 10/2020. DOI:10.1017/ice.2020.488

2. Gibson K, Mody L, Sax H, **Gontjes K**, Cassone M, Mang N, Wolfensberger A: Understanding the Short-Term Dynamics of MRSA Between Patients and Their Immediate Environment. *Infection Control and Hospital Epidemiology*.41(S1): s420-s420, 10/2020. DOI:10.1017/ice.2020.1077

## Software / Apps

1. Gontjes K. ISScreener. Snakemake workflow to run insertion sequence detection on Illumina sequencing data. 01/2024. <https://github.com/kylegontjes/ISScreener>
2. Gontjes K. GeneScreener. Snakemake workflow to quickly screen for a query sequence in a subject sequence. 01/2024. <https://github.com/kylegontjes/GeneScreener>
3. **Gontjes K**. phylokit. Snakemake pipeline to generate a phylogenetic tree from a whole-genome alignment. 01/2024. <https://github.com/Snitkin-Lab-Umich/phylokit>
4. Gontjes K. CoverageStats. Snakemake workflow to calculate coverage statistics after mapping reads to a reference genome. 01/2025. <https://github.com/kylegontjes/CoverageStats>
5. **Gontjes K**. phyloAMR. R package for phylogenetic analysis of antimicrobial resistance. 01/2025. <https://github.com/kylegontjes/phyloAMR>

## Preprints

1. **Gontjes KJ**, Singh A, Sansom SE, Boyko JD, Smith SA, Lautenbach E, Snitkin E: Phylogenetic context of antibiotic resistance provides insights into the dynamics of resistance emergence and spread, 06/2025. PM40502560
2. **Gontjes KJ**, Pirani A, Badri Narayanan D, Lojek LJ, Han JH, Tolomeo P, Goldstein EJ, Lautenbach E, Snitkin E: Identifying drivers of  $\beta$ -lactam/ $\beta$ -lactamase inhibitor resistance emergence and spread before their clinical deployment, 07/2025. PM40766120
3. Boyko JD, **Gontjes KJ**, Snitkin ES, Smith SA: Uncertainty in joint Ancestral State Reconstruction: Improving accuracy and biological interpretability of ancestral state prediction, 10/2025. PM41279858
4. Thiede SN, Steinberg H, Benedict E, Sansom S, Aroutcheva A, **Gontjes KJ**, Winner K, Royzenblat S, Pirani A, Weinstein RA, Zawitz C, Moore NM, Popovich KJ, Snitkin ES: Antibiotic-resistance plasmid amplified among MRSA cases in an urban jail and its connected communities, 10/2025. PM41282926

## Abstract/Posters

1. Gontjes K: Qualitative & Quantitative Assessment of Multidrug-Resistant Organism Transmission Potential of Rehabilitation Equipment in Skilled Nursing Facilities, Abstract, Annual Meeting for Society for Healthcare Epidemiology of America, Boston, MA, 2019
2. Gontjes K: Multidrug-Resistant Organism Burden of Common Area & Rehabilitation Environment in Skilled Nursing Facilities, Oral Presentation, University of Michigan Department of Internal Medicine 27th Annual Research Symposium, Ann Arbor, MI, 2019
3. Cassone M, Lansing B, Mantey J, Gibson K, **Gontjes K**, Mody L: 917. Persistence of Multidrug-Resistant Organisms during Occupancy Changes in the Nursing Home Setting, and Impact of Patient Hand Hygiene Assistance, *Open Forum Infectious Diseases*.7(Supplement\_1): s493-s493, 2020. DOI:10.1093/ofid/ofaa439.1105
4. **Gontjes KJ**, Gibson K, Lansing B, Mantey J, Jones K, Cassone M, Wang J, Mills J, Mody L, Patel PK: 156. How Does Exposure to C. Diffogenic Antibiotics Impact Multidrug-resistant Organism Colonization and Environment Contamination in Nursing Homes?, *Open Forum Infectious Diseases*.7(Supplement\_1): s207-s207, 2020. DOI:10.1093/ofid/ofaa439.466
5. Gibson K, Mody L, Sax H, Gontjes K, Cassone M, Mang N, Wolfensberger A: Understanding the Short-Term Dynamics of MRSA Between Patients and Their Immediate Environment, Abstract, Poster, Virtual6th International Conference on Healthcare Associated Infections, 2020
6. Gontjes K: Simplifying Surveillance Screening: Can Environmental Surveillance Replace Perianal Screening?, Oral Presentation, IDWeek 2020, Virtual, 2020

7. Cassone M, Lansing B, Mantey J, Gibson KE, **Gontjes KJ**, Mody: Impact of Hand Hygiene Assistance and Room Occupancy Changes in Persistence of Multidrug-Resistant Organisms in Nursing Homes, Poster, Abstract, AGS 2022, Orlando, FL, 2022
8. Cassone M, Lansing B, Mantey J, Gibson KE, **Gontjes KJ**, Mody L: Diversity and Persistence of MRSA and VRE in Nursing Homes: Environmental Screening and Whole Genome Sequencing, SHEA Spring 2022, Colorado Springs, CO, 2022
9. **Gontjes K**: Carbapenem-resistant *Klebsiella pneumoniae* Non-susceptibility to Two  $\beta$ -lactamase Inhibitor Combination Therapy Agents Before Their Market Introduction, Poster, NHGRI Professional Development Conference, Salt Lake City, UT, 2023
10. Gontjes K: Carbapenem-resistant *Klebsiella pneumoniae* Non-susceptibility to Two  $\beta$ -lactamase Inhibitor Combination Therapy Agents Before Their Market Introduction, Oral Presentation, ASM Microbe, Houston, TX, 2023
11. Gontjes K: Application of phylogenetic context to patient risk factor analysis illuminates features associated with the emergence and spread of antibacterial resistance, Poster, University of Michigan Department of Internal Medicine Research Seminar, Ann Arbor, MI, 2024
12. Gontjes K: Application of phylogenetic context to risk factor analysis illuminates patient features associated with the emergence and spread of antibacterial resistance, Oral Presentation, Kleb Club Seminar, Virtual, 2024
13. Gontjes K: Genomic and Patient Features Associated with Non-Susceptibility to Meropenem-Vaborbactam & Imipenem-Relebactam Before Their Market Introduction, Poster, Lake Arrowhead Microbial Genomics Conference, Lake Arrowhead, CA, 2024
14. Gontjes K: Application of phylogenetic context to patient risk factor analysis illuminates features associated with the emergence and spread of antibacterial resistance, Oral Presentation, University of Michigan Infectious Disease Scholarship Symposium, Ann Arbor, MI, 2025
15. **Gontjes K**: The role of genetic background in the emergence and spread of antibiotic resistance in carbapenem-resistant *Klebsiella pneumoniae*, Oral Presentation, Michigan Infectious Disease Genomics (MIDGE) Center's Annual Meeting, Ann Arbor, MI, 2025
16. Gontjes K: Identifying drivers of  $\beta$ -lactam/ $\beta$ -lactamase inhibitor resistance emergence and spread before their clinical deployment, Oral Presentation, Midwest Microbial Pathogenesis Conference, St Charles, IL, 2025
17. Gontjes K: Identifying drivers of  $\beta$ -lactam/ $\beta$ -lactamase inhibitor resistance emergence and spread before their clinical deployment, Oral Presentation, National Institute of Allergy and Infectious Diseases (NIAID) Genomic Centers for Infectious Diseases (GCID) Annual Meeting, Virtual, 2025
18. **Gontjes K**: Ancestral state reconstruction for studying the emergence and spread of clinically important traits, Oral Presentation, Michigan Infectious Disease Genomics (MIDGE) Center's Annual Meeting, Ann Arbor, MI, 2026